

Received 10 January 2026, accepted 25 January 2026, date of publication 3 February 2026, date of current version 6 February 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3660002

TOPICAL REVIEW

Safe Search and Rescue Operations Based on Autonomous Robots: A Systematic Review of the General System Architecture

ABBAS AQEEL KAREEM^{1,2}, AHMED JABBAR ABID¹,
DALAL ABDULMOHSIN HAMMOOD², (Member, IEEE),
ALI ABDULELAH AL-NAJI^{1,3}, (Member, IEEE), AND JAVAAN CHAHL^{3,4}

¹Electrical Engineering Technical College, Middle Technical University, Baghdad 10022, Iraq

²Technical Engineering College of Artificial Intelligence, Middle Technical University, Baghdad 10022, Iraq

³School of Engineering, Adelaide University, Mawson Lakes, SA 5095, Australia

⁴Platforms Division, Defense Science and Technology Group, Edinburgh, SA 5111, Australia

Corresponding author: Dalal Abdulmohsin Hammood (dalal.hammood@mtu.edu.iq)

ABSTRACT Over the past few years, remarkable progress has been made in Autonomous Robotic Systems (ARSS) due to their multidisciplinary nature and applications. Among the most in-demand applications is the Search and Rescue (SAR) task, particularly in high-risk environments where human entry is unsafe. Specifically, teleoperated SAR platforms deploy dedicated sensors and control algorithms to optimize search effectiveness and mitigate risks incurred by human responders. This systematic review analyzes 74 research papers published between 2016 and 2024, with a focus on the role of robots in SAR operations (sensor integration), the types of robots employed, and the control algorithms used. The work classifies robotic systems in the context of SAR as either aerial, ground, or marine platforms and analyzes their sensor technologies, autonomous navigation functions, and Human-Robot Interaction (HRI) capabilities. A major contribution of this survey is the proposal and design of a five-layered architecture for robotic SAR systems, which categorizes system functions into clearly differentiated layers; namely, the HRI Layer, Perception Layer, Navigation Layer, Control Layer, and Robot Layer. Such a structured method enables modular, scalable, and real-time decision-making under dynamic disaster conditions. Moreover, this paper identifies several research challenges, including multi-robot coordination, Artificial Intelligence (AI) based real-time decision-making, and energy efficiency, providing a related analysis for the future development of robotic SAR systems.

INDEX TERMS Robotics, search and rescue (SAR), multi-robot search and rescue, autonomous robots, disaster robotics, search and rescue robots, systematic review, UAVs, drones.

I. INTRODUCTION

Geological hazards generate tens of thousands of fatalities, displace millions, and cause huge losses for infrastructure each year, on average, due to earthquakes throughout the globe. Between 2016 and 2024 (the period during which this review was conducted), approximately 3616 natural and 1405 technological disasters occurred [1]. In particular,

The associate editor coordinating the review of this manuscript and approving it for publication was Alba Amato¹.

Fig. 1 illustrates that the phenomenon can be caused by both natural and technological disasters, exhibiting distinct characteristics in different years and regions. Specifically, more cases are observed in Africa and Asia compared to other regions of the world, particularly in recent times. Additionally, there is a slight surge in the number of events around 2020, with some regions experiencing a peak. In this regard, systematic and organized measures against the hazard can significantly reduce disaster damage risks. For example, in September 2018, a 7.5-magnitude earthquake and

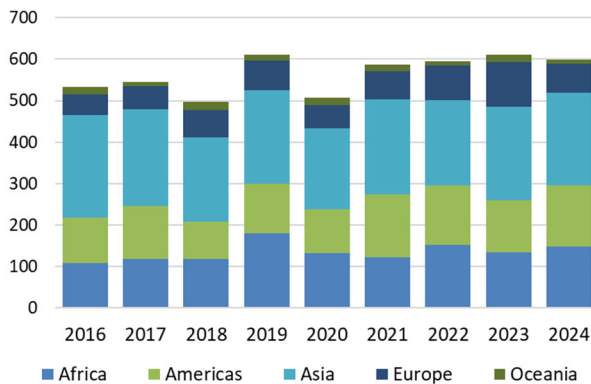


FIGURE 1. Disaster occurrences during the years, based on data from the Center for Research on the Epidemiology of Disasters [1].

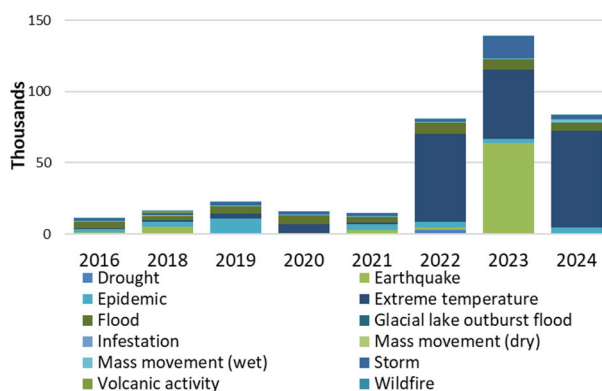


FIGURE 2. Number of deaths from natural disasters by type, 2016 to 2024 [3].

subsequent tsunami struck Palu and Donggala in central Sulawesi, Indonesia. While the number of deaths was comparatively small (4,340), the earthquake ended up displacing over 200,000 people and destroying or damaging over 40,000 homes [2]. As a result of power and communications lines being cut, many residents did not receive tsunami warning messages. Additionally, in the twenty-first century, millions of people have died due to various types of natural disasters worldwide, as reported in Fig. 2 [3].

To minimize the loss of life caused by disasters, effective Search and Rescue (SAR) operations are essential. Therefore, the number of deaths has decreased in recent years, despite the increase in the number of disasters recorded, as shown in Fig. 2, due to technological developments and modern methods in disaster management operations and SAR of survivors.

Disaster response encompasses many time-sensitive actions aimed at achieving a core objective; namely, identifying and locating survivors to maximize life-saving efforts. According to previous studies, the survival rate is 91% one hour after a disaster occurs. After the first day, the percentage drops to 81%. After the second day, the percentage reached 36.7%. By the fourth day, the rate reaches 19% [4]. On the other hand, in some cases, it is difficult for humans

to enter the affected areas. In this context, time and safety are two crucial factors in SAR missions [5]. Therefore, the search process must be as quick as possible to preserve the lives of survivors and, at the same time, save the lives of the SAR team from risks as much as possible [6]. To handle such issues, an effective combination of safety and efficiency could be achieved using Robotic Search And Rescue (RSAR) systems [7]. Over the last few years, several advancements have been made in robot-assisted SAR operations. This is in particular the case when robots are utilized, since the time to search and find permissible survivors is faster [8], [9]. Particularly, robots can help reduce the difficulty of SAR work in dangerous and contaminated environments, improve the chances of victims surviving, and save the lives of the SAR members [10], [11]. Software robots can also process a time map and surveil a disaster area [12], [13].

After a meticulous search of the literature, including the review papers, 11 documents were found. The role of robots, Artificial intelligence (AI), and sensors in the face of natural disasters is undergoing extensive revisions, ranging from water to air. The authors of [14] also claimed the absence of a systematic survey on different computer vision techniques pertinent to the requirements associated with successive phases of flood management and propose a common taxonomy for mapping them. Moreover, the authors presented possible computer vision technologies and challenges as potential solutions for the future. However, the review is limited to flood management, excluding other disasters, and lacks validation through real case studies and full integration into disaster response systems. In addition, Unmanned Aerial Vehicles (UAVs) in civilian applications were extensively reviewed in [15]. Civil applications of UAVs covered in this paper include road traffic monitoring, providing wireless coverage, remote sensing, SAR, delivery/unmanned courier services, security/surveillance systems, precision agriculture, and civil infrastructure inspection. The research also highlighted the major challenges of UAVs used in all application domains, such as charging and autonomy challenges, obstacle avoidance and swarming issues, network/computer vision, and security-related challenges. As a general survey, this work lacks an in-depth review of specific UAV applications, technical issues, and comparisons with other technologies. Moreover, the authors of [16] focused on the applications of UAVs in RSAR, highlighting the benefits of UAV systems and sensor use in SAR, while defining tactical considerations. Additionally, they highlighted issues such as energy constraints, inclement meteorological conditions, restrictive government policies, and the need for real-time data analysis that hinder the development of other robotic platforms. While the study provides several examples of UAV-based SAR, it does not offer any comparisons with alternative search strategies that utilize ground robotics, thereby limiting an assessment of relative effectiveness. In [17], the authors gave an overview of multi-robot SAR and provided a detailed discussion of major technical challenges that prevent the deployment of swarm systems within SAR operations. The survey pointed

out several open problems, such as scalability, real-time perception, network robustness, etc., as well as the gap between simulated and real robotic performance. While the work provides theoretical understanding and results from a simulation, there is a lack of validation from real-world experiments or empirical measurements and benchmarking against pre-existing single robot SAR methodologies. Consequently, this is a limitation that affects the practical evaluation of multi-agent SAR systems. In another work [18], the importance of trustworthy artificial intelligence in natural disaster management, including risk assessment, prediction, and response, was explored. The authors classified the AI algorithms for disaster applications by six disaster categories and reviewed the evolution from machine learning to Deep Learning (DL), as well as data fusion and explainable AI. However, a detailed technical discussion on the accuracy of models and their comparative performance in disaster management is lacking. Furthermore, the importance of sensors in the field of disaster management and SAR operations was studied in [19], where their usefulness in detecting survivors and assessing disaster conditions was emphasized. Particularly, the integration of assistive technologies, for example, AI, Internet of Things (IoT), UAVs, and wireless sensor networks, to improve the efficiency of sensors in SAR missions was discussed [19]. Moreover, the authors in [20] proposed a systematic review of ground disaster survivor detection methods during the introduction of critical issues, such as the lack of real-world validation and high-power consumption. Specifically, the review focused on multi-sensor fusion for enhanced robustness, TinyML-based computation, and computation deployment optimization. Although the authors assumed that networks (Wi-Fi, LTE, LoRa, BLE) are available, in real disaster situations, these infrastructures are likely to be damaged and out of order. The authors of [22] also discussed recent IoT and sensor networking technologies for SAR operations in flooded areas, focusing on their role in detecting survivors and disaster response in real-time. They discussed different communication protocols, including LTE, 5G, Wi-Fi, and ad hoc networks, and reported on the main challenges related to each of these, such as energy consumption, infrastructure requirements, and interoperability. The work proposed an IoT-based approach for flood mitigation based on the exploitation of heterogeneous multi-sensor data and wireless networks to enhance the SAR effectiveness. In [21], the authors conducted a systematic review of DL applications, especially semantic segmentation networks, for natural disaster management, focusing on forest fires, floods, and earthquakes. They studied Convolution Neural Network (CNN) based networks such as U-Net, SegNet, and DeepLab for disaster impact assessment and identified issues in their implementation; namely, the absence of real-world validation, feature redundancy loss, and AI interpretability. The paper demonstrated multi-sensor integration, as well as AI-based prediction, as significant progress and calls for scalability and practical use in an exemplary manner. Additionally, the authors in [22] also investigated the use

of Digital Twins (DT) for post-disaster risk management (PRM) with a view to their potential for improving disaster response, allocating resources optimally, and furthering recovery. The work presented an improved disaster prevention and reduction management theory (DPRMT) that can be used to assist decision-making, predict casualties, and assess resilience. Nevertheless, the work hardly explains how DT can be integrated into creative emergency response systems, including creative countries and humanitarian organizations. A survey of multi-robot SAR systems was presented in [7]. The authors investigated task allocation, path planning, AI-based perception, and reinforcement learning to improve SAR operations in urban, maritime, and wilderness environments. However, the human-AI teamwork and capacity to work in challenging conditions have not been fully developed.

Table 1 summarizes the differences and novelty of our study in relation to prior reviews, highlighting a broader range of coverage that includes multifaceted aspects and diverse technologies in AI, sensors, and robotics. Current studies are mostly isolated to specific parts of the RSAR systems, such as sensor-centric solutions or AI-based perception elements, without a systemic study of system-level integration. Noticeably, the key parameters that are relevant for the deployment in the real world are mostly neglected in previous works, such as computational type (i.e., cloud, edge, and fog computing), testing environments (real world versus simulation), and Human-Robot Interaction (HRI). Moreover, few reviews have consistently critiqued the interrelationship among multi-robot coordination, AI techniques, and sensor technologies on the SAR process. This survey fills these gaps in a broad, interdisciplinary manner by establishing a strong new position. In particular, the main contributions of this survey can be summarized as follows:

1. Proposing a five-layer structure for RSAR systems.
2. Considering full integration of AI, sensors, and robotics in RSAR systems comparison.
3. Comparing multi-robot and single-robot approaches to SAR.
4. Highlighting research gaps and discussing avenues for future work to build NextGen RSAR systems across various natural disaster settings.

II. METHODOLOGY

This section outlines the step-by-step process that is followed to conduct a Systematic Literature Review (SLR) of the literature related to the investigation of RSAR operations, using a methodical framework that seeks to reduce potential sources of bias. The current work follows Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines in order to ensure precision and completeness [25]. Accordingly, a preliminary set of research questions was considered in order to guide the following review of RSAR missions. Specifically, a thorough search strategy was used to identify pertinent studies with the search string refined using an iterative process of trial and error

TABLE 1. Comparative analysis of prior SAR surveys in relation to the present study.

Survey	Path Planning Algorithms	Single & Multi Robot	AI Techniques	Sensor Type	Robot Type	Human-Robot Interaction (HRI)	Computational Type	Testing Environment
[14]	No	No	Yes	Yes	No	No	No	Yes
[15]	No	No	No	Yes	Yes	No	Yes	Yes
[16]	Yes	No	Yes	Yes	Yes	No	No	No
[17]	Yes	Yes	Yes	No	Yes	Yes	No	Yes
[18]	No	No	Yes	No	No	No	No	No
[24]	No	No	Yes	Yes	Yes	No	No	No
[20]	No	No	Yes	Yes	Yes	No	Yes	Yes
[25]	No	No	No	Yes	No	No	Yes	No
[21]	No	No	Yes	Yes	No	No	No	No
[22]	No	No	Yes	No	No	No	Yes	Yes
[23]	Yes	Yes	Yes	No	Yes	No	Yes	No
*	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

* Means this work

in order to achieve a sensitivity that was as optimal as possible for the RSAR-related literature. Particularly, identified studies were filtered using a pre-specified set of criteria for the exclusion process following a thorough screening of titles, abstracts, periodicals, and complete texts. A standardized extraction instrument of data was then developed, which included general bibliographic information (such as the name of the publisher) and specific variables of interest, such as AI algorithms, robot typologies, and sensor modalities. Finally, the extracted data were analyzed and categorized using empirical conventions.

A. RESEARCH QUESTIONS

This research employed four Research Questions (RQs) to achieve the study's objectives. These RQs were identified to enable a more thorough investigation of the state-of-the-art methods within this research area, identifying key advances and potential deficiencies. More precisely, the SLR methodological analysis is based on the following RQs:

RQ1: How are robot systems being utilized in SAR operations?

RQ2: What are the categories of robots used for SAR work?

RQ3: What types of remote sensing systems are used in the search operation?

RQ4: To what extent can Machine Learning (ML) methods aid SAR autonomy?

B. INFORMATION SOURCES AND SEARCH STRATEGY

To have a robust literature, the present SLR retrieved relevant literature from two major core collections, including Scopus and Web of Science (WoS). In particular, these databases were chosen because they index many peer-reviewed journals and conference papers, providing comprehensive coverage of SAR robotics development attempts. Other resources, such as ScienceDirect, IEEE Xplore, and Springer, have also been

coordinated to enhance the databases. In this regard, the search strategy included Boolean searches using the related keywords. Additionally, a manual search was also performed on Google Scholar. A total of 1,156 articles were initially retrieved from all databases (Scopus, WoS, ScienceDirect, IEEE Xplore, and Springer). The inclusion criteria were English-language publications from 2016, and the search was limited to 2024. The chosen period (2016–2024) was selected to account for the latest developments in RSAR and place them in a proper historical context, within which to assess emerging trends. Fig. 3 illustrates the yearly distribution of papers found across all digital libraries.

C. SEARCH CRITERIA

Search criteria are defined following the guidelines in [26], applying them individually to the RSAR systems. To find the best possible matches among literary works, specific keyword searches and their synonyms are utilized. Primary studies have been selected through a two-stage process, involving the search for keywords connected by proper Boolean expressions (i.e., AND/OR). In the first part, all the studies related to SAR systems are explored and shortlisted, while studies of RSAR systems are identified in the second part. Regarding SAR systems, the search strings created and modified to meet the advanced search requirements of each database are: (“search and rescue” OR “SAR”) AND (“robot” OR “UAV”) AND (“deep learning” OR “multi-agent system”). All keywords were queried in the databases mentioned, since it is the leading scientific source database. All retrieved records were fully screened without applying any stop criterion. Furthermore, search filters were used, such as publication year, language, and article type.

D. INCLUSION AND EXCLUSION CRITERIA

The retrieved papers were manually screened to select the relevant studies in the first phase of the search. A series of Exclusion Criteria (EC) was considered in accordance with the study's objective. Papers that were not written in the English language or for which the full text was not available were excluded. Furthermore, we excluded works that were not concerned with RSAR operations or only briefly mentioned the RSAR system, as well as works that did not make direct contributions to the SAR system. In addition, theoretical studies that were not practically implemented or empirically verified were also removed to concentrate on developed RSAR systems.

E. SELECTION OF SOURCES

In this SLR, a two-step process is followed in the selection of a specific article:

1. Screening based on title and abstract.
2. Selection of the most relevant articles based on a full-text analysis.

Initially, sample keywords are applied to the electronic search databases to ensure that the eligibility criteria are met. A relatively large number of articles and research studies

are listed, including all publishers in that database. As a first step, a screening process is conducted among the shortlisted articles and research studies based on their titles and abstracts to identify the relevant studies. In contrast, the remaining articles are excluded from the review. In the second step, the articles collected from the first step are reviewed to assess their relevance to the objectives of the secondary research. Inclusion/exclusion criteria are ensured during both steps.

F. DATA COLLECTION PROCESS AND ANALYSIS OF RESULTS

The following section defines the data collection process for the RSAR subsystems. In this regard, two different data-extraction tables have been developed to extract relevant information from the selected articles for each subsystem. Specifically, the attributes considered in the first table are: (i) Reference, (ii) Design issues, (iii) Method, (iv) Techniques, and (v) ML/DL algorithms. The attributes chosen for the second table are: (i) Reference, (ii) single or multi-robot systems, (iii) Experiments/Simulations, (iv) sensor types, (v) computational type, and (vi) path planning. These attributes have been used to collect and efficiently summarize the information related to RSAR sub-systems. The reviews of the shortlisted articles are presented separately for each subsystem (detection/tracking) to facilitate a more specific and thorough analysis.

G. BIBLIOMETRIC ANALYSIS OF THE RETRIEVED PAPERS

The literature, collected from all available databases, included publications from January 2016 to May 2024, as shown in Fig. 3. A discernible escalation in the output of research was found from around the year 2021, and this was subsequently amplified in preceding years. This ascending trend indicates the growing relevance and recognition of RSAR technologies, especially in the context of global emergencies, natural catastrophes, and the growing need for autonomous robotic systems in operational real-world SAR missions.

We conducted a bibliometric analysis based on authorship, keywords, and journal publication patterns. For each study, we extracted the database source, journal title, article title, index terms/keywords, authorship information, citation counts, and the year of publication. Subsequently, a word cloud was created to indicate the relative frequencies of the identified keywords. In Fig. 4, the most common keywords are written in a larger font, while less frequent terms are written in a smaller font. Keywords were grouped based on semantic similarity, and the most frequent keywords were: deep learning (202 occurrences), unmanned aerial vehicles (UAV) (148), aerial vehicle (114), aircraft detection (98), and object detection (86). Collectively, these are the key terms that support the idea that modern RSAR research is increasingly turning towards aerial detection systems using artificial intelligence, and these will cover applications as varied as the detection of human victims or broader object localization in complex environments.

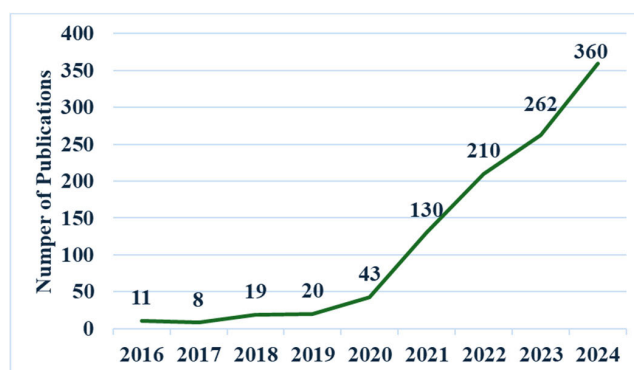


FIGURE 3. Yearly distribution of the papers found in digital libraries.

In addition, a co-occurrence analysis is done to outline key trends and technologies in the world of RSAR. This method of analysis structures keywords based on their relations to each other, as displayed in a visualization in Fig. 5. More specifically, the figure highlights three main thematic trends: the red cluster shows advances in the field of DL, the green one in the field of object detection, the blue one in the field of UAVs and their integration with path, remote sensing, and machine learning techniques, and the purple one on outcomes related to search and rescue systems. In addition, the brown cluster represents the achievement of full autonomy and adaptive environment capacities in drone solutions.

The inter-cluster connections shown in Fig. 5 highlight some of the technological modalities used for the acquisition of disaster and survivor data. For example, the keyword “lidar” is linked to object identification and image processing; the terms “path planning” and “remote sensing” are correlated with UAVs, which suggests the potential for victim reporting; the phrase “computer vision,” and “multi-UAV” and “attention mechanism” co-occur with UAV-based search and rescue operations. Core tasks are therefore reduced to the keywords “search and rescue”, “multi-UAV”, “computer vision”, and “deep learning”.



FIGURE 4. Word cloud for the most frequently used keywords.

Fig. 6 is a thematic map based on the co-word analysis of Fig. 5, which splits keywords into four quadrants based on their stage of development and relevance. The upper right quadrant is labeled “Motor topics”, the lower right

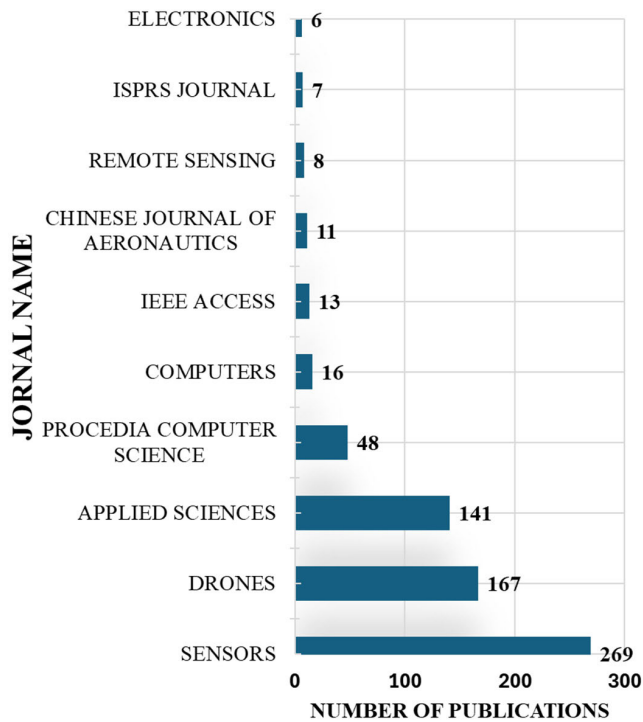


FIGURE 8. Top ten active publishers.

H. DEMOGRAPHIC DATA OF SLR

In this part, we discuss the implementation of the review protocol for information and data collection. Particularly, we followed the two-step study selection process outlined in the study, where articles were initially screened based on title and abstract, resulting in the inclusion of 150 articles and the exclusion of 1006 studies. This high number of excluded articles is due to the exclusion criteria mentioned. The same exclusion criteria were also applied to the second step of the study selection process, where the articles were screened again based on the full-text reading. More specifically, the contents of the screened articles from the first step were studied and reviewed, and they were excluded only when any of the exclusion criteria were satisfied. In conclusion, the 74 articles shortlisted for analysis and critical review concern SAR operations. The overall scheme of the employed review protocol is depicted in Fig.10, according to the flow diagram proposed in [25].

Once the primary articles for the survey were selected, each work was reviewed by two authors, and the third author also reviewed it in case of any conflict or disagreement between the authors.

III. RESULT

A. DOCUMENT QUALITY SYNTHESIS

A document quality assessment process was performed during the full-text analysis to provide an overview of the study’s methodology and practical validation. The scoring was independently performed by two researchers to minimize bias. The quality evaluation consisted of two main criteria rated from 1 (lowest) to 3 (highest), with the final score being

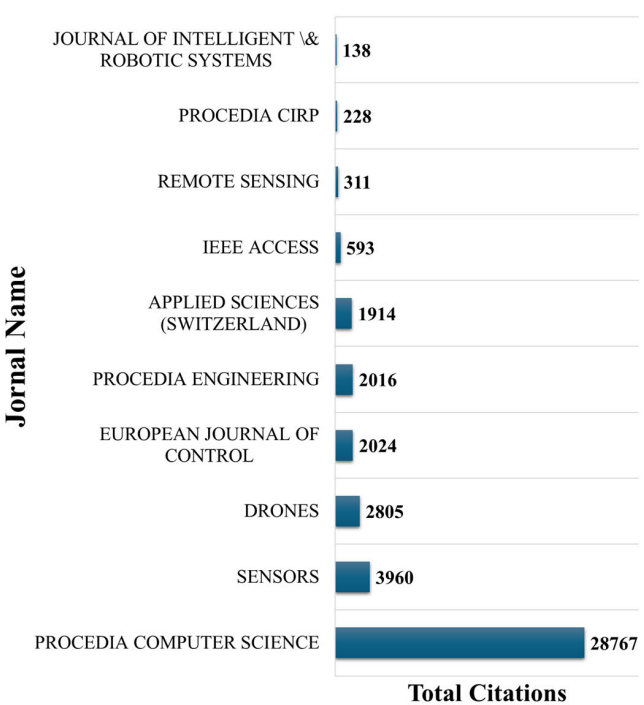


FIGURE 9. Top ten cited publishers.

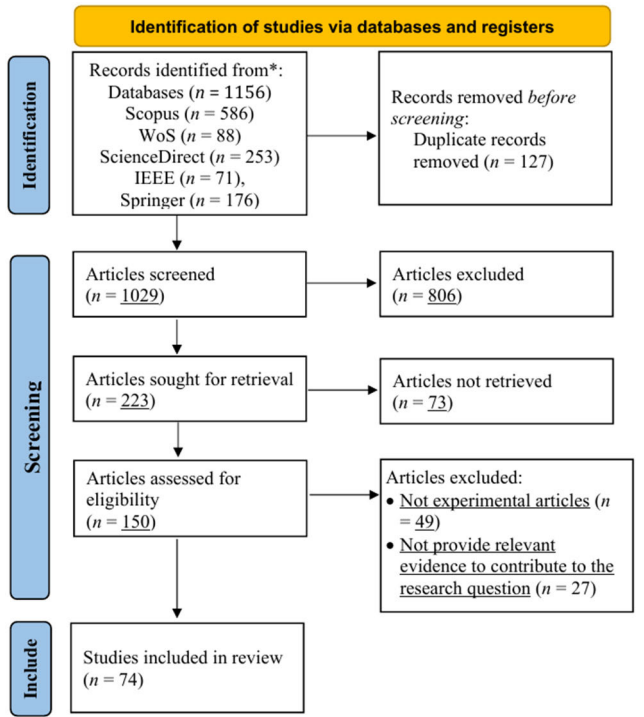


FIGURE 10. PRISMA diagram for the identification, selection, and inclusion processes.

the sum of both scores. The scoring was given based on the following metrics:

Validation: Simulations have a score of one point, simulations that have partial real-world validation receive two points and fully real-world get three points.

Methodology: Minimal reproducibility and fewer implementation details receive one, High reproducibility and more implementation details receive two, and robust documents with reasonable transparency and limitations acknowledged score three.

The mean quality score from all 74 documents considered for full-text analysis was 3.6 points, with a standard deviation of 1 (Fig. 11). Documents with a score of 4 or higher are considered high-quality documents. Thus, 43 documents (58.1%) are considered high-quality documents, providing an overall good sample count and a robust methodology. Accordingly, we conclude that research in this field is in a mature phase, where methodologies are well-supported by previous studies and sample sizes are high, which can be justified by the current growth potential of robots in the SAR field.

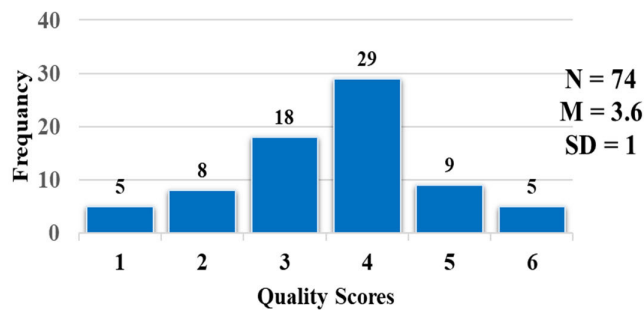


FIGURE 11. Quality assessment scores histogram.

To assess the degree of experimental realism across the included studies, we analyzed the validation settings used in each study. As illustrated in Fig. 12, a slight majority of studies (55%) utilized real-world testbeds, while 45% relied on simulation environments.

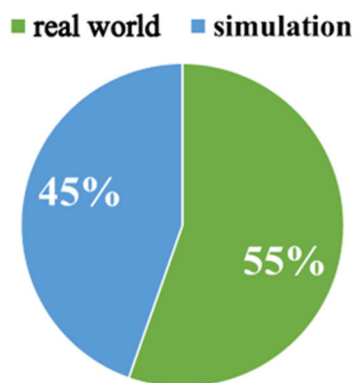


FIGURE 12. Proportion of real-world and simulation-based validation.

B. EVALUATION OF STRENGTHS AND WEAKNESSES IN THE RSAR SYSTEM

To provide a clear insight into the most recent achievements, this section will compare the included papers in this SLR, which were published from 2016 to 2024 (Fig. 13) across

aerial, ground, and hybrid platforms. By considering the scope, strengths, and limitations of each study, this paper provides insight into what can currently be offered in terms of capabilities, design compromises, and operational challenges for real-world rescue.

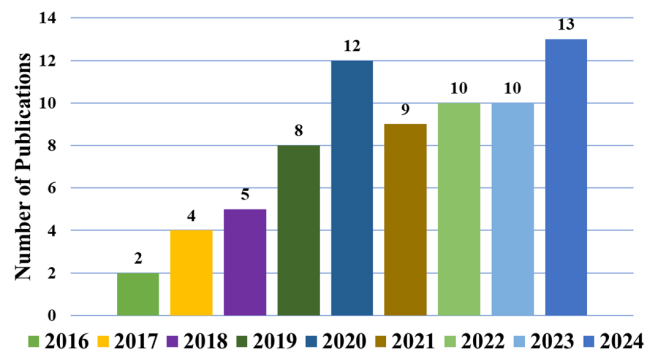


FIGURE 13. Yearly distribution of the included papers in SLR.

The reviewed studies encompass a wide range of technological approaches, including flying robots (e.g., UAVs equipped with thermal, visual, or multispectral sensors), Unmanned Ground Vehicles (UGV) platforms, and solutions that integrate airborne and ground-based equipment. Various papers highlight the utilization of onboard artificial intelligence, including CNNs, reinforcement learning, and ensemble models, for real-time victim detection, obstacle avoidance, and mission adaptation.

Operationally, the potential benefits included better spatial coverage (swarms of agents coordinated to search a rugged area more efficiently), improved detection robustness (using one or several types of sensors for features like thermal-visual, thermal-gas, or acoustic-visual fusion), autonomy in crowded spaces and Global Positioning System (GPS) denied regions, and cost-effective implementations with lightweight or embedded systems (e.g., CNN inference on Raspberry Pi). There are also a few experiments that demonstrate the scalability of the modular system architecture and the Robot Operating System (ROS) based framework, which can be deployed at a large scale.

Nevertheless, consistent limitations are observed. These include ambient environmental conditions (heat noise, low visibility, or vegetation occlusion) sensitivities, impaired ability to detect non-cooperative or occluded victims in need of rescue, high computation and energy consumption rates, as well as the increasing difficulty to maintain an inter-robot communication and synchronization process, particularly in decentralized or swarm-based systems. Many researchers also require annotated datasets and markers such as AprilTags or assume ideal terrain to limit generalization in the real world. In addition, many solutions are still not validated in the field but only tested under controlled conditions, making them less ready for real disaster use.

Table 2 systematically records the generic contributions of a wide range of SAR-centred research works. Each bullet

point describes the main focus of the research, including thermal-RGB sensor fusion for real-time onboard AI processing, modular snake robots, and swarm-based coordination strategies, as well as the corresponding operational advantages, such as enhanced coverage, fast victim localization, or the ability to operate in cluttered environments. The demonstrated limitations are also equally stressed, which include hardware limitations, occlusion sensitivity, environmental dependency, and scale-dependent issues. This broad comparison provides significant insight into the trade-offs faced in modern RSAR system design, leading to further studies that yield robust, efficient, and deployable solutions for disaster response situations.

C. MULTI-LAYERED ARCHITECTURE FOR RSAR SYSTEMS

An orderly, layered architecture is essential in RSAR operations to maintain loose coupling of functions, thereby allowing for system modularity and decreasing complexity. The modularity of SAR is achieved through separate layers of operating functionalities, which allow each layer to be independently developed and optimized for better integration, thereby significantly increasing the success rate in task running, sensor installation, and decision-making.

This organized manner also makes scalability and flexibility easier for complex, dynamic disaster environments, which makes it particularly applicable. In RSAR systems, several multi-layered architectures have been proposed to enhance efficiency and autonomy. Among the most interesting architectures, the Aerostack Multi-Layered Approach presented in [98] was first designed for aerial robotic applications. Aerostack employs a layered control architecture that comprises a deliberative layer (mission planning), an executive layer (task management), a control layer (real-time execution), and hardware abstraction layers for sensors and actuators. This architecture can facilitate modular system integration, although it lacks an obvious mechanism for multi-robot coordination. Additionally, sensor perceptiveness and navigation functionality are intertwined, which may constrain its real-time adjustability during search and rescue operations. Another example of multi-layer architectures is the one in [99], authored by Hierarchical Multi-Robot Planning Framework. In this case, a task splitting approach among the different robots can be modeled, utilizing a global layer for mission allocation and local planning to delegate specific subtasks to individual robots. Although this method leads to better cooperation among multi-agents, it lacks a systematic sensor integration mechanism that is clearly defined within the framework. Furthermore, the lack of an HRI layer means it is less suitable for applications that require human assistance and interaction.

To compensate for these downsides, here we propose a five-stacked SAR structure with various important modifications. First, the latter includes a dedicated HRI layer to enable real-time human-in-the-loop interventions, supervisory control, and collaborative decision-making.

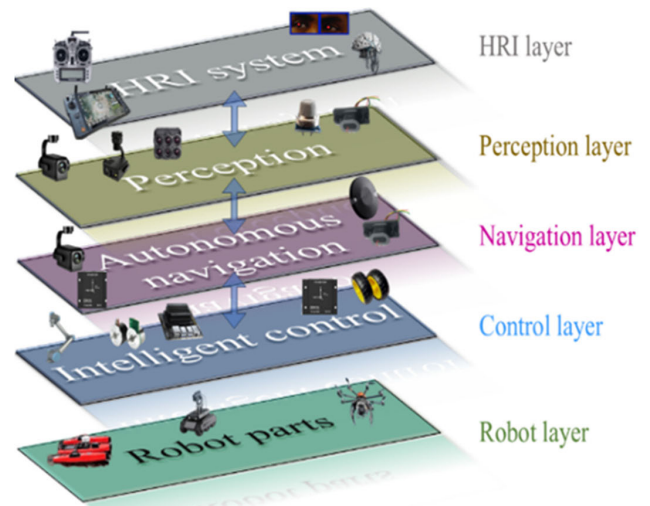


FIGURE 14. Multi-layered architecture for robotic SAR system.

Unlike existing frameworks, it can differentiate between perception and navigation and further perform multi-sensor fusion for improved environmental awareness and AI-based autonomous mobility.

Moreover, the framework enhances multi-robot coordination, enabling UAVs, Unmanned Ground Vehicles (UGVs), and Unmanned Underwater Vehicles (UUVs) to collaborate on SAR missions. Not only does this layered style decrease the complexity of implementation, but it is also more easily scalable, modular, and efficient than any previous design when applied to SAR tasks. The workflow of the SAR system is proposed in Fig. 14 and comprises five individual layers, which perform the tasks to be fulfilled for searching:

1) HRI LAYER (HUMAN-ROBOT INTERACTION)

This layer encompasses communication interfaces, control systems, and human supervision in SAR operations. It is essential for enabling human-robot symbiotic autonomy and co-decision making. Normally, this is done through Remote Control (RC) and laptop control/monitoring. At times, brain signals and eye movements can even be used to guide robots. By providing operators with real-time situational awareness and control through remote interfaces, this layer ensures that humans can supervise missions from a safe distance. In addition, emergency overrides and intuitive teleoperation interfaces further minimize risk.

2) PERCEPTION LAYER

This layer integrates various sensors (LiDAR, cameras, radar, thermal imaging, and depth sensors) for environmental perception, obstacle detection, and victim localization. The perception layer enables robots to perceive their environment through the use of cameras, sense their environment with touch sensors, and also detect odors using gas sensors. Through onboard sensors, this layer enables autonomous detection of victims and environmental hazards, thereby reducing the need for human scouts.

TABLE 2. The advantages and limitations of the reviewed RSAR papers.

Ref	Focused area	Advantages	Limitations
[28]	Integration of thermal and optical cameras on UAVs for real-time survivor detection	Provides rapid localization using onboard AI; effective thermal-based detection	Limited in dense occlusion scenarios; prone to false positives
[29]	Optical sectioning to detect humans under the forest canopy	Works under partial occlusion; robust to lighting; autonomous control	Depends on accurate altitude; high processing load
[30]	Multi-UAV coordination in maritime SAR	Wide-area coverage; effective communication protocols	Sensitive to weather conditions; complex UAV coordination
[31]	3D UAV path planning using terrain knowledge	Efficient path generation; minimizes overlap	Assumes access to high-quality 3D maps; limited real-time adaptation
[32]	Ethical and design framework for SAR robotics	Offers policy insights; identifies design bottlenecks	Lacks empirical validation; limited technical evaluation
[33]	Behavior-based cooperative control in multi-UAV SAR	Decentralized autonomy; responsive to dynamic environments	Scalability issues; requires task-specific tuning
[34]	Onboard ML processing for SAR UAVs	Real-time processing; flexible hardware; open-source	Limited by onboard compute power and energy usage
[35]	Thermal-RGB sensor fusion with AI-based survivor detection	Operates in low visibility; fast inference models	False detection under heat noise; motion blur challenges
[36]	Heavy ground robot with enhanced mobility and payload	Robust in rough terrain; supports heavy equipment	Difficult to maneuver in tight spaces; semi-autonomous
[37]	UAV-UGV collaboration for mapping and victim detection	Combines aerial overview and ground precision; map sharing	Requires synchronization; risk of communication delays
[38]	Sensor fusion for victim detection	Combines multiple spectral sources for improved accuracy; effective in diverse conditions	High computational demand; limited real-time capability in complex terrains
[39]	Real-time object detection using YOLOv3	Fast inference; good performance in cluttered environments; tested on real disaster datasets	Accuracy drops with occlusion and low lighting; fixed threshold settings may reduce adaptability
[40]	Modular snake robot for confined space navigation	Flexible movement in debris-filled environments; adaptable to various terrains	Limited payload capacity; slow movement in open areas
[41]	Vision-based detection system for persons in SAR	Fully automated workflow; adaptable to aerial and ground systems	False positives in visually cluttered scenes; detection range limited by resolution
[42]	Embedded vision system on UAV using unsupervised learning	No need for labeled data; fully autonomous; reduced training complexity	Lower accuracy compared to supervised methods; sensitive to noise and scene variation
[43]	Hybrid algorithm for indoor SAR detection	Effective for indoor victim detection using combined methods; improved robustness	Algorithm complexity and runtime are not ideal for real-time applications, as they depend on the indoor layout
[44]	Thermal imaging for tracking in drone-based SAR	Reliable in low-light and obscured conditions; continuous tracking	Affected by non-human heat sources; less effective in high ambient temperatures
[45]	Aerial SAR using real-time YOLO-based object detection	Fast image processing; drone integration for SAR missions; scalable system	Dependent on drone altitude and camera resolution, challenges with object scale
[46]	DL-based drone imagery analysis for SAR	High sensitivity detection; scalable with drone swarms; cost-efficient using commercial drones.	Delayed detection due to offline processing; limited resolution in live feed
[47]	Low-cost thermal victim detection using CNN on UAV	Cost-effective deployment; real-time detection using a simple CNN on Raspberry Pi; compact system.	Limited detection distance (~3 m); performance drops under occlusion or in low-contrast thermal scenes
[48]	Victim detection in landslides using thermal and gas sensors	Integrates gas (CO₂) detection and thermal imaging; tests in real landslide-like environments.	Sensor fusion is not fully automated; camera mounting/angle variability affects results
[49]	SAR via multispectral image anomaly detection	Robust anomaly detection using PCA and color clustering; suitable for large-scale aerial imagery.	Not tested on actual victim cases; false positives possible in natural background variation
[50]	Victim detection in thermally cluttered backgrounds	Combines RGB and thermal with entropy maps to suppress heat clutter; YOLOv4 for final detection.	High computational complexity; needs accurate entropy mapping; potential false alarms in overlapping thermal regions
[51]	Thermal detection using deep CNNs and UAVs	Real-time person detection; tests with YOLOv4-tiny and Faster R-CNN; validated in real scenarios.	Reduced accuracy under heavy occlusion; thermal noise in variable sunlight environments
[52]	Multi-modal victim localization using layer-based maps	Uses semantic 3D mapping + thermal data; improves localization by reducing false positives.	Requires accurate sensor fusion; complexity of layered map generation
[53]	Exploration + semantic segmentation for efficient victim search	Combines SLAM and multi-class segmentation; improves victim detection rates.	Requires ample computational resources; sensitive to poor lighting conditions
[54]	Dual-modal sensing (RGB + thermal) for outdoor victim detection	Bimodal fusion enhances robustness; it uses object orientation estimation and segmentation.	More effective in day scenarios; night detection under vegetation needs further improvement
[55]	Debris-based victim detection with ResNet-50 and decision tree classifiers	High accuracy (99.5%) with random trees; low computational cost; robust under occlusions	Works on a custom RGB dataset; not validated on real UAV scenarios; not fully autonomous
[56]	Coordinated UAV swarm (thermal, visual, and bio-radar) for SAR	Distributed DL-based detection + radar confirmation + medical delivery UAV	Requires functional coordination of three UAV types; limited scalability without a strong communication infrastructure
[57]	Multi-stage Deep Reinforcement Learning (DRL) + heuristic planning for search missions	Combines DRL with an informed search for better convergence and task allocation	Complex training may be computationally intensive in real-time UAV platforms
[58]	Comparative analysis of exploration strategies in simulation	Provides a benchmark for multiple SAR strategies	Limited to simulation; lacks real-world sensor/environmental noise
[60]	Operational assessment of drone-based search in the wilderness	Real-case deployment with analysis of visual capability and area coverage	Manual search analysis; no autonomous detection or AI integration

TABLE 2. (Continued.) The advantages and limitations of the reviewed RSAR papers.

[61] Feasibility review of UAVs in mountainous SAR	• Covers challenges of elevation, temperature, and payloads	• Theoretical; lacks hardware-in-the-loop testing
[62] Object tracking in SAR via visual appearance features	• Lightweight algorithm; suitable for embedded UAV systems	• Visual-only tracking; fails in poor lighting or occlusion scenarios
[63] Drone-based life signs detection (e.g., respiration/motion)	• Non-contact vital sign sensing via radar/sensors	• Susceptible to motion artifacts; affected by distance and angle
[64] Swarm robotics for distributed search	• Scalability: cooperative behavior improves coverage	• Communication overhead and synchronization complexity
[65] Layered Search and Rescue (LSAR) algorithm with multi-UAV	• Focuses on survivor clustering in the disaster center; outperforms auction-based and LIAM approaches	• Centralized coordination needed; requires cloud or server infrastructure
[66] Human detection via voice recognition from UAV-mounted microphones	• Detects victims hidden from camera (in shadows/obstacles); fuses vision + sound	• Rotor noise interference; performance degrades with distance; requires an array mic setup
[67] Camera-based detection and GPS-based positioning for SAR	• Lightweight, accurate position estimation using ArUco markers and a simple architecture	• Requires GPS signal and visual line-of-sight for marker detection
[68] CNN-based thermal image classification for avalanche victim detection	• High-accuracy CNN classifier, real-time classification on embedded GPU	• Needs training data for avalanche-specific scenarios; computational load
[69] IoT-based indoor victim detection with audio processing	• Human voice-based detection; deployable in buildings	• Relies on sound emission from victims; poor performance in silent environments
[70] PIR and thermal sensor-based human detection robot	• Low cost; real-time detection; simple sensor integration	• Limited detection range; may not work accurately in high-temperature environments
[71] Autonomous victim search using ROS, YOLO, Light Detection and Ranging (LiDAR), and AprilTags	• Modular architecture; scalable and replicable using standard ROS packages	• AprilTag dependence for victim recognition; may misclassify or miss untagged humans
[72] Thermal vision with YOLO-based recognition in SAR	• Robust thermal-based human recognition; effective in dark/smoky environments	• A thermal-only approach may confuse non-human heat sources
[73] TurtleBot3-based exploration and victim localization using AprilTags	• Fully autonomous SLAM with victim detection using information-theoretic exploration	• AprilTag dependency; indoor-only operation; limited field deployment proven
[74] Swarm UAV simulation for SAR using visual victim detection	• Coordinated UAV swarm planning; vision-based simulation for SAR	• Limited to simulation; lacks real-world deployment validation
[75] Smart Rescue Robot for SAR	• Integrates multiple sensors (thermal, gas, sound); can operate autonomously in hazardous zones	• Limited terrain mobility; lacks robust obstacle avoidance in complex environments
[76] Ensemble Learning for Victim Detection	• Ensemble models improve detection accuracy; tested on real SAR conditions	• Real-time performance unverified; energy demands for multiple classifiers
[77] Snake-Inspired Robot for Earthquake SAR	• Snake-bot design enables access to narrow/confined spaces	• Complex control system; limited speed in wide-area search
[78] Detection of Dead Victims using AI	• Uses CNNs on aerial images for victim classification; focuses on cadaver detection	• Relies on image quality and annotated datasets; unsuitable for live victim localization
[79] Deep Learning for Victims in Cluttered Urban Areas	• Utilizes Mask R-CNN with UAV imagery; robust to clutter and occlusion	• High dependency on model training; may misclassify non-human shapes in rubble
[80] Mixed-Reality Quadruped Robot for Guidance	• Combines mixed-reality interface with legged robots for precise navigation	• Deployment cost is high; it requires a human operator for the interface
[81] Enhanced UGV for SAR with Multimodal Sensors	• Improved terrain handling; combines thermal, gas, and image data	• Weight and power consumption limit operational time
[82] Embedded Autonomous SAR Mobile Robot	• Fully embedded system with real-time detection; portable platform	• Not scalable for large-area operations; low-level processing power
[83] Semi-Autonomous Mobile SAR Robot	• Combines manual override with autonomous mode; multi-sensor fusion	• Partial autonomy requires human-in-the-loop; communication loss can disrupt the mission
[84] GAN-based Audio Noise Suppression	• Enhances microphone signal clarity using GANs in noisy environments	• Only applicable to audio cues; don't detect victims directly
[85] Mobile robot using ultrasonic & PIR sensors for indoor SAR	• Cost-effective, real-time, work in confined spaces	• Limited outdoor use, lacks vision/classification
[86] Snake-like robot for navigating rubble/collapsed areas	• Highly flexible, fits narrow spaces	• Slow, fragile, lacks detection integration
[87] AI prioritization using thermal data in fire scenarios	• Victim State Prioritization, thermal logic	• Simulated only, lacks real-world testing
[88] Swarm-based coverage planning with multiple drones	• Fast coverage, team coordination	• Requires strong comms, lacks fine detection
[89] YOLO-based indoor navigation & detection robot	• Works without GPS, accurate detection	• Indoor only, not scalable outdoors
[90] Real-time YOLOv5 model for victim & object detection	• High speed, good accuracy	• Needs visibility, poor in debris/darkness
[91] Manually controlled thermal robot for SAR	• Good low-light performance, terrain adaptable	• Not autonomous, limited coverage
[92] Multimodal fusion UAV for better human detection	• Illumination-robust, higher accuracy	• High model complexity, hardware load
[93] Low-cost autonomous UAV using PIR + obstacle sensors	• Autonomous, GPS-aided, simple design	• Sensor limits detection precision
[94] Aerial victim detection using a multi-copter drone equipped with visual and thermal sensors	• Uses multiple sensor types to enhance detection accuracy	• Limited performance in GPS-denied indoor environments and relies on high-altitude imaging, which may reduce resolution
[95] I, including speed, the required load, and the harsh environment in which the robot operates, and motion sensors	• Integrates multiple low-cost sensors	• Lower accuracy compared to advanced vision systems

TABLE 2. (Continued.) The advantages and limitations of the reviewed RSAR papers.

[96]	Development of a fully decentralized UAV swarm using Particle Swarm Optimization (PSO) to coordinate exploration across vast and unknown environments	• Fast convergence to target areas using PSO- Effective coverage of unknown environments	• Dependent on reliable inter-UAV communication for coordination
[97]	Indoor UAV-based search and detection using a monocular camera, LiDAR, and CNN-based object detection	• Real-time autonomous navigation in GPS-denied cluttered indoor environments	• Monocular depth estimation lacks robustness compared to stereo or RGB-D setups.
[98]	Use of a snake-like robot with serpentine locomotion for detecting life signals in cluttered rubble environments during urban SAR operations	• Multi-sensor fusion (sound, temperature, motion) for victim detection	• Limited real-world testing- Accuracy of detection not evaluated at scale

3) NAVIGATION LAYER (AUTONOMOUS NAVIGATION)

This layer includes path planning algorithms, obstacle avoidance, and localization techniques to enable autonomous movement in dynamic environments. Moreover, this layer supports multi-robot coordination and adaptive mobility strategies for different terrains. Additionally, this layer contains all the sensors used for the robot's autonomous navigation, including GPS, an RGB camera, a laser, and an Inertial Management Unit (IMU). Path planning and obstacle avoidance algorithms ensure safe, autonomous traversal of complex terrains, preventing accidents that might otherwise endanger human rescuers.

4) CONTROL LAYER (INTELLIGENT CONTROL)

This layer governs low-level control mechanisms, motor actuation, and stability management. In addition, it implements adaptive control algorithms for precise motion execution in unpredictable conditions. More specifically, this layer provides real-time responsiveness and stability to the robot's movements. Fail-safe mechanisms (e.g., return-to-home and loss-of-signal protocols) are implemented here to ensure mission continuity without putting humans at risk.

5) ROBOT LAYER (PHYSICAL SYSTEM)

This layer consists of mechanical and hardware components, including UAVs, UGVs, legged robots, and modular robotic structures. It ensures the structural integrity, power management, and mobility capabilities of SAR robots. The physical platform itself is designed to operate in dangerous conditions. Redundant power systems, protective frames, and robust communication links ensure operational reliability without relying on human proximity.

The SAR system was designed using a five-layered approach, following data synthesis of the included studies in this review. After conducting a thorough investigation of the available SAR paradigms, we demonstrated that there was a need for an approach that can provide system organization, reduce complexity, and enhance the modular features of the RSAR systems. By decoupling the functionalities of SAR, perception, navigation, control, and HRI, they can be optimized independently of each other to provide better scalability, adaptability, and real-time decision-making in a disaster environment. Although the five-layer structure

facilitates the systematization of RSAR functions, its actual performance is largely determined by the extent to which it can be integrated into SAR protocols, hardware limitations, and mission-centric constraints. Future work should focus on real-world validation, computational efficiency optimization, and the robustness of autonomous decision-making in challenging scenarios.

D. ROBOTS TYPE USED IN SAR MISSIONS

This part categorizes RSAR into three main types: Aerial (UAVs), Ground (UGVs), and Marine (UUVs). Additionally, emerging hybrid SAR systems that integrate multiple robot types are discussed. This classification is based on recent research, technological advancements, and real-world SAR deployments.

1) AERIAL ROBOTS

UAVs are one of the most widely employed robots in SAR operations, due to their speed and agility in covering large areas effectively [100]. In practice, UAVs can be deployed rapidly to disaster-stricken areas and provide real-time visual and thermal images for victim detection [101]. In this regard, UAVs for the SAR system can primarily be categorized into two types. The first one is Fixed-Wing UAVs that have been designed for long-range surveillance, and they are well-suited for wide-area search (i.e., post-earthquake reconnaissance). However, they are dependent on runways or launch systems and are not capable of hovering [102]. The second type of multi-rotor UAVs can hover, fly under clutter, and detect victims at close range. Their operating time is, however, reduced due to their shorter battery life. This type is more easily controlled in complex terrains with obstacle, and is more flexible in dealing with strong winds [103]. The list of advantages and disadvantages for each type of UAV is presented in Table 3 [104], [105].

2) GROUND ROBOTS

UGVs are key equipment for SAR and are particularly used in complex environments, such as earthquake debris, collapsed buildings, and underground areas. In particular, UGVs in SAR can be divided into three main types [106]. The first one is tracked UGVs, which are known for their high maneuvering capabilities (they can move efficiently through

debris and rubble), but consume significant energy [107] “Ferret”, For instance, in just 100 hours, the researchers removed all workers from the ruined mine [108]. The second type is legged UGVs, which share the advantages of better stability and a simpler structure, and can also traverse a diverse range of terrains. Over the past few years, there has been a growing interest in this class of robots, which play a critical role in enhancing the efficacy of SAR operations, yielding significant improvements through the utilization of such technology [109]. Quadruped robots are capable of dynamically changing their locomotion to smoothly avoid obstacles in an unstructured environment [110]. The wheel robot is also a third category of ground robots. This one is faster-paced, and we can use it among research team members to support our communication. It is capable of climbing over obstacles in rough areas [111]. They are 3-, 4-, and 6-wheel robots, depending on their use (speed, load to carry, and the environment in which the robot is used) [112]. Such a robot is the most popular because it is used in many applications and is relatively simple to design. Since legged and wheeled robots share similarities, their advantages and disadvantages are summarized in Table 4 [109].

TABLE 3. Types of fixed and multirotor UAVs with their advantages and disadvantages [104], [105].

	Fixed wing	Multi-rotor
Advantage	• Higher flight safety • More power consumption. • Long flight duration. • Perfect stability.	• Ability to take off and land vertically. • The landing/ take-off substation area is not required • Ability to hover at one point. • Ability to perform agile maneuvering.
Disadvantage	• Does not have hovering capability. • Need a large area for take-off and landing.	• Low speed and short flight range. • Small payload capacity.

3) MARINE ROBOTS

UUVs play a critical role in water-based SAR operations, including flood rescues, underwater searches, and maritime disaster response. Marine robots have proven their ability to carry out SAR operations in a way that increases the probability of the operation being successful [113]. The integration of renewable energy sources, such as solar-powered or wave-energy harvesting systems, has enabled UUVs to conduct long-duration operations [114]. This makes it one of the most widely used types of robots in the field of searching for missing persons underwater, drawing marine maps, or finding lost objects in the seas [115].

An emerging topic in the RSAR system is the design of multi-type robots for a hybrid SAR system, aiming to enhance efficiency in disaster relief. Unlike single-robot SAR

systems, multi-robot SAR systems can leverage the complementary features of aerial robots, ground robots, and marine robots to conduct a more thorough, real-time, and adaptive search and rescue operation. These systems incorporate collaborative autonomy, sensor fusion, and AI-enabled coordination to operate more efficiently, respond more quickly, and reduce the level of human risk in hazardous areas. To this end, one of the well-investigated multi-robot SAR schemes is UAV-UGV cooperation, in which the role of UAVs is to quickly explore large disaster-struck regions and provide feedback to UGVs for their intervention. One of the most frequently studied multi-robot SAR applications is the collaboration between UAVs and UGVs, in which UAVs are employed for quick scanning/surveillance over extensive disaster areas, providing feedback to UGVs operating on-site. For example, UAVs equipped with LiDAR, RGB-D cameras, and a thermal vision system can be used to locate survivors, recognize damaged building images, and develop 3D maps of the disaster environment. The maps are then sent to UGVs, which can travel independently through rubble and other fragile terrain in debris fields to perform close-in searches, deliver first aid supplies, or aid in evacuations. This interoperability significantly reduces the time it takes for SAR teams to respond and assist victims. For example, in [116], the authors developed an AI UAV-UGV system, in which the UAVs perform aerial surveillance/Home detection, and a UGV, working in conjunction with reinforcement learning, moves autonomously toward the survivors. Beyond aerial-ground coordination, there is also an increasing focus on integrating UGVs and UUVs, such as in flood disasters, maritime vessels, and underwater rescue operations, including sonar, water LiDAR, and littoral robotic UGV data. Find-swale robots are used at disaster sites to measure water depth, detect underwater obstacles, and search for survivors. In these missions, UGVs serve as extraction points for UUVs, which are then set free to perform accurate underwater search tasks.

TABLE 4. Legged and wheeled robot advantage AND disadvantage [109].

	Legged	Wheeled
Advantage	• Long run time (~60 min). • Payload: up to 50% of the platform weight. • Can handle uneven ground terrain. • 2D Omnidirectional Movement. • No impact from wind disturbances.	• Long run time (~180 min) • Payload: up to 150% of the platform weight. • No impact from wind disturbances.
Disadvantage	• Cannot handle vertical terrain. • Absence of motion in the z-direction.	• Absence of motion in the z direction. • Can only traverse relatively even surfaces. • Can have difficulties maneuvering in tight spaces • Cannot handle vertical terrain

The deployment of multi-robot SAR systems enhances disaster response; however, it has several issues related to coordination, communication, energy efficiency, and flexibility. When there are many robots, centralized control quickly becomes infeasible because of computational and communication limitations. The use of decentralized AI-driven coordination in real-time decisions relating to swarm behavior is fundamentally important because it provides such agents with decentralized AI-based coordination, which is only possible with the assistance of swarm intelligence and multi-agent reinforcement learning (MARL). Dynamic disaster settings require adjustments to task allocation to account for the failures of robots or changes in the environment. AI-based workload distribution can also provide better resilience by enabling robots to automatically redeploy tasks with minimal human intervention. Moreover, the system faces severe communication constraints due to destroyed infrastructure and signal interference. 5G/6G-based UAVs or ad hoc mesh networking/satellite applications do facilitate better connectivity, even though it is still challenging to deliver low-latency and high-bandwidth communication in energy-limited scenarios. Although robots tend to be more autonomous, HRI is still necessary for the command and monitoring of a robot. In this context, Augmented Reality (AR) interfaces, gesture-based controls, and Brain-Computer Interfaces (BCI) enhance operator situational awareness and intervention efficiency. Based on the articles collected, as shown in Fig. 15, we note that UAV usage accounted for 88% and UGV usage for 12% in 2019. In 2020, the integration of multiple robot types was proposed for the SAR process. For example, the integration of UAVs and UGVs was implemented in a single SAR system at an 8% rate. The integration of UGV with UUV is also at a rate of 8%. This year marks the first year in which research began integrating multiple robot types to conduct search and rescue operations. In 2022, the percentage of integration between the UGV and UAV increased to 10% of the total research published that year. In 2024, marine robots were used at a rate of 8% in search and rescue operations, according to the research published that year.

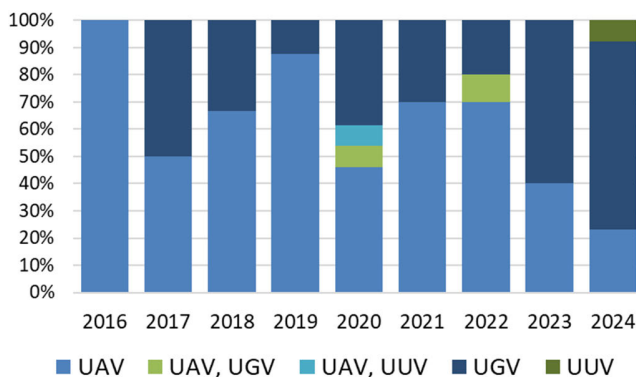


FIGURE 15. Usage of Robot types in SAR operations.

Indeed, the dominance of UAVs as platforms for SAR operations is underpinned by several unique operational benefits. In particular, UAVs offer better airborne agility, quick on-demand deployment, and long-range line-of-sight capabilities, which become highly desirable for time-sensitive missions over unreachable, dangerous, or rapidly changing areas. Furthermore, UAV-based platforms are inherently suited to lightweight sensor systems and have relatively low launch preparation requirements, ensuring they can be deployed quickly in the field with limited logistical support. Their designs also enable them to be integrated with more sophisticated autonomous systems, such as on-the-fly decision-makers and adaptive path planners. In comparison with UGVs and UUVs, which can be limited by terrain complexity or environmental occlusion, UAVs can operate more flexibly in various environments, ranging from dense urban areas to wide-area wilderness. Their overall exploitation is therefore an expression of and a response to the technical and operational needs in modern SAR conditions.

E. SENSORS TYPES USED IN RSAR SYSTEM

Through modern sensors, it is possible to overcome environmental challenges and enhance search and rescue missions using robots. For example, an aerial photo from high altitudes can scan a larger area and explore faster if good analysis algorithms are used [116]. Moreover, through acoustic sensors, the efficiency of finding survivors can increase, especially in remote areas [117]. Therefore, sensors are considered one of the most critical components in designing a robotic SAR system. In this respect, sensors can be divided according to the principle of operation into photoelectric sensors, electromagnetic sensors, acoustic sensors, and chemical sensors, such as gas sensors [118], [119], [120]. Several types of sensors can be used in both navigation and perception layers, a section of which will be given below:

1) VISUAL SENSORS (VS)

One of the most essential types of sensors is those that use optical input in search and rescue operations, where RGB cameras are used to detect objects. Additionally, a 3D RGB-D camera was used to scan objects and achieve higher accuracy in detecting objects in the search and rescue process [121]. When there are no vision capabilities at night, for example, researchers resort to using security cameras in search and rescue operations. Thermal cameras can determine the location of the target with high accuracy, as they produce thermal fingerprints that can be distinguished from the background, and also detect structural defects [28].

2) ACOUSTIC SENSORS (AS)

Acoustic sensors can be used in different environments and can cover vast areas by detecting sound frequencies and vibrations. The omnidirectional microphone can also sense audible frequencies and ultrasonic waves in the air, as well as underwater [122]. This type of sensor can be used

to determine the source of sound and its direction with greater accuracy. In addition, this sensor can be used to determine the source of human footsteps or to identify people who are swimming, boats, or submarines [123]. However, one of the disadvantages of this type is the noise present in the environment, which can cause a percentage of error in SAR operations. Likewise, the presence of obstacles reduces the strength of the received signal, which increases the error rate in detecting and determining the source of the sound [124].

3) BIOLOGICAL SENSORS (BS)

Biological sensors can detect any element that originates from or is related to biological materials. This type of sensor is a controversial example in SAR operations. Numerous biological elements are involved in SAR operations for humans, animals, and plants. There are many types of biological sensors, such as those for detecting human odor or pathogens in agricultural samples [125]. These sensors can be utilized in robots to locate survivors and victims. Biological sensors are often integrated with robots to facilitate various SAR operations [126]. One of these operations involves equipping a group of dogs with an electronic nose equipped with sensors that detect the scent of their surrounding environment. The air contents indicate whether or not there are living humans. There are other applications for searching inside a poisoned environment or discovering unexploded poison bombs, etc. [127].

4) CHEMICAL SENSORS (CS)

Chemical sensors are an essential class of sensors in SAR operations. These types of sensors can be used to detect toxic gases and hazardous materials in various environments, including secret laboratories, as well as during SAR operations. It also includes the detection of chemical warfare agents, toxic industrial chemicals, and environmental pollutants [128]. However, the challenges in this type of sensor lie in integrating this chemical device with robotic devices, which increases the complexity of system design in SAR operations using robots, which will increase the system's cost and take longer than other methods [129].

5) NAVIGATION SENSORS (NS)

Navigation sensors are used in SAR operations, as they provide a comprehensive description of the victim's location and direction [130]. In particular, the navigation sensor is one of the devices that cannot be used in SAR operations, especially in cases where there is no GPS [131].

Multi-sensor fusion in SAR is crucial for enhancing detection probability and reducing the false alarm rate, and has the potential to improve decision-making in real-time. Sensor fusion refers to the process of aggregating data from multiple sensors to leverage their complementary capabilities and mitigate individual drawbacks. In the literature, the practical applications of multi-sensor fusion for SAR operations have been demonstrated recently. For instance, the authors of [7]

presented a collaborative multi-robot SAR system, where LiDAR, thermal imager, and RGB streams are combined to enhance environmental perception and victim detection. In a related work [20], the application of multi-sensor technology was investigated for ground-based disaster response, with sensor fusion employed to enhance detection accuracy while minimizing computational load. Studies for UAV-based SAR systems (e.g., [56]) have shown that the presence of UAVs/devices equipped with LiDAR sensors and thermal cameras enhances search processes, which in some cases may be possible in a larger space when ground access is minimal.

Conversely, in [24], the efficiency of seismic sensors and RGB-D cameras for terrain analysis and victim localization in cluttered environments was highlighted. In marine SAR operations in [22], sonar and acoustic/optical sensors were investigated to enhance the efficiency of underwater search operations in the context of flood rescue missions. These examples demonstrate the increasing importance of sensor fusion for urban SAR applications, where robots are expected to operate relatively autonomously in dynamic disaster environments. Nevertheless, open questions persist regarding the real-time synchronization of multi-sensor data, efficient AI-based data processing, and energy-efficient integration mechanisms.

As shown in Fig. 16, we can see that visual sensors are primarily employed in 57.831% of SAR operations, indicating their efficiency and accuracy in completing tasks. They are followed by the fusion of visual and navigation sensors, with 18.072%, which ranks second, indicating a significant integration between these two sensors in carrying out SAR missions. application of a triad of navigation sensors-optical sensor-chemical sensor, with 13.253%. This variety utilizes multiple sources to reach victims, enabling SAR operations in diverse regions. The use of navigational sensors exclusively, and the use of acoustic and optical sensors in combination, account for 5%. As for acoustic sensors, they were used in 1% of all research found during this review.

The independent but complementary sensing characteristics of these sensors explain why thermal and RGB visual sensors are used in most RSAR platforms. Lower limb images are acquired using thermal imaging that operates in the Long Wave InfraRed (LWIR) spectrum, which allows for the passive detection of human thermal signatures under occluded or low-illuminating conditions. This makes it indispensable for nocturnal, obstructed daylight, occluded, and debris-covered environments. On the other hand, RGB cameras can provide high-resolution context awareness, which is essential for semantic segmentation, visual object tracking, and object classification solutions, particularly when combined with convolutional neural networks (e.g., U-Net, DeepLab) or real-time detectors (e.g., YOLOv5). This bimodal fusion creates robustness by complementing the shortcomings of each sensor type, where thermal sensors lack spatial details, and RGB sensors are sensitive to low light. UAV payload capabilities, power budget constraints, and mission trade-offs between coverage and resolution also limit the selection

of the sensor. For thermal-RGB fusion, real-world deployment prefers thermal-RGB free integration due to its robustness in diverse environments and its suitability for integrating with on-board AI inference pipelines to implement victim detection.

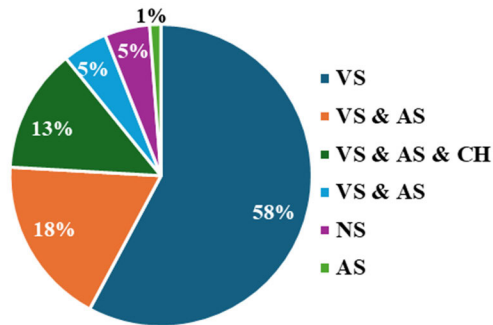


FIGURE 16. Usage proportions of sensor types.

F. HUMAN-ROBOT INTERACTION (HRI) IN RSAR MISSION

The introduction of HRI in small team leader roles during a SAR mission is crucial to enhancing efficiency and safety. In practice, successful HRI enables human first responders and robots to collaborate with minimal friction, thereby maximizing performance in hazardous environments. Therefore, the quality of human-robot interaction is a critical component of the success of such missions. In fact, effective HRI can help robots understand human commands, respond to humans in a timely and proper manner, and gain better situational awareness, knowledge, and decision support [132]. HRI definition in SAR can be divided according to autonomy and control:

1) TELEOPERATION (REMOTE CONTROL)

In teleoperation, human operators directly interact with robotic systems to control navigation, object manipulation, and environmental monitoring in real-time. Such a method provides fine control, but may be hindered by communication latencies and operator load [133]. Examples of recent teleoperated robotic SAR include the Earthshaker robot, designed to operate in complex disaster scenarios [134]. Specifically, the Earthshaker is a track platform equipped with a robotic arm and a gripper, utilizing various sensors and controllers to operate in hazardous conditions, including debris removal and victim retrieval. Through teleoperation, human operators can remotely command/control the robot to perform tasks accurately and perceive them in real-time, which improves the safety and efficiency of SAR missions.

2) SUPERVISORY CONTROL

Supervisory control systems enable human operators to monitor a group of autonomous robots, issuing high-level commands and intervening when necessary. Contrary to teleoperation, this method minimizes the operator burden

and allows for semiautonomous decision-making by the robot [135]. It is particularly advantageous in a catastrophic disaster on a large scale, where robot teleoperation per se is not feasible. In this regard, a significant application of supervisory control was demonstrated in the DARPA Subterranean (SubT) Challenge, which involved multi-robot teams exploring subterranean environments to locate survivors and identify hazardous conditions. In this task, operators controlled mixed-mode robots, including aerial (drones), ground-based (wheeled and tracked) platforms, and legged robots, to search caves, tunnels, and underground facilities as autonomously as possible. The system focused on high-risk zones, allowing controllers to intervene where and when required. The DARPA Subterranean (SubT) Challenge used a multi-modal task-oriented Graphical User Interface (GUI) to coordinate groups of robots with mixed modalities in subterranean environments. More precisely, the DARPA user interface provided a 3D map view, which was updated in the SLAM dovetail with real-time updates, interactive positioning of task markers, and prioritization regions, enabling the controller to define tasks easily. Decentralized decision-making was also made possible by a multi-robot task allocation framework, which reduced the workload of the operator. Furthermore, audio cues enhanced operators' situational awareness by providing them with real-time status, allowing operators to manage robots effectively without relying solely on visual feedback. Situational awareness, productivity, and adaptive real-time control were enhanced in these time-critical SAR missions by this system.

3) COLLABORATIVE CONTROL

Collaborative control can be a more flexible and dynamic HRI, in which the decision making is shared between human(s) and robot(s), while being coordinated in a time-critical way. This method is based on self-perception and reasoning about action, with human intervention, for cases in which robots encounter neutral or critical situations. Multi-modal communication systems (e.g., voice commands, gesture recognition, and augmented reality interfaces) are frequently employed to support natural interaction [136]. The Interactive Human-oriented Exploration and Supervision Under Scarce Communication (iHERO) approach is one of the most recent examples using human-robot collaboration in SAR tasks [137]. This system addresses the problem of poor communication in underground environments and, finally, but not least, human-in-the-loop interaction with autonomous robot fleets. The iHERO framework enables operators to update on the progress in exploration, re-focus on regions/areas, and dynamic adaptation based on instructions. Robots in such a system are capable of exploring unknown environments independently and cooperating with the operator on demand; they update their status to the operator as the mission progresses. Detailed simulations and prototype experiments have verified the practical potential of iHERO in real SAR systems, facilitating efficiency and safety in challenging scenarios.

Effective communication with robots is one of the problems that confronts the researchers in designing an HRI. It is also a challenge to keep the robot's work environment in mind. The trade-off between automatic control and manual intervention is also one of the complex problems that occupies much more attention in this perspective [138].

G. COMPARATIVE ANALYSIS

Robots have been increasingly recognized as a key component in SAR operations due to their capability of fast reaction, high situational awareness, and enhanced life-saving effects during disasters. They encompass a broad and heterogeneous variety, ranging from single-robot SAR missions, where one robot carries out the search task entirely autonomously or semi-autonomously, to multi-robot systems that involve several coordinated robots working together to achieve improved efficiency and coverage. In this respect, it is essential to compare and contrast these two SAR techniques to determine which one yields better results. In this section, key contributors to evaluating both single-robot and multi-robot SAR systems are discussed. The comparison is based on the following key values:

1. **Path Planning Algorithms:** Classifying manual, semi-autonomous, and autonomous navigation schemes.
2. **Computational Architecture:** Considering cloud computing, Personal Computer (PC)-based computing, and embedded-processing techniques.
3. **Testing Environments:** Distinguishing between the simulation and SAR experimentations.
4. **Sensor Technologies:** Categorizing the kinds of sensors used for victim identification, navigation, and environment sensing.
5. **Robotic Platforms:** SAR robots can be classified as UAVs, UGVs, and UUVs.
6. **Detection Algorithms:** Investigating the role of AI algorithms (e.g., YOLO, SSD, and CNN models) for object and survivor detection.

This review encompasses 74 thoroughly reviewed research papers, with 60 focused on single-robot SAR systems and 14 devoted to multi-robot SAR systems. These reports were systematically selected and carefully examined for additional helpful information. As evident from Fig. 17, the research articles have been published in various countries. Seven of which are the United States, China, Spain, Japan, the United Kingdom, Australia, and India, which reflects global interest in the development of the SAR technology. The topic was most popular in India and Spain, with seven papers from each. China was next with six papers. Japan sponsored four research papers in the area of robots used for SAR, ranking third. The remaining countries, including Arab nations and others, also showed interest in this subject, with one or two published research papers from each.

Looking at the testing spaces, found in Table 5, we notice a very strong predominance of the category "Real world", which amounts to an impressive eight references, which indicates a strong tendency or dependency on testing models

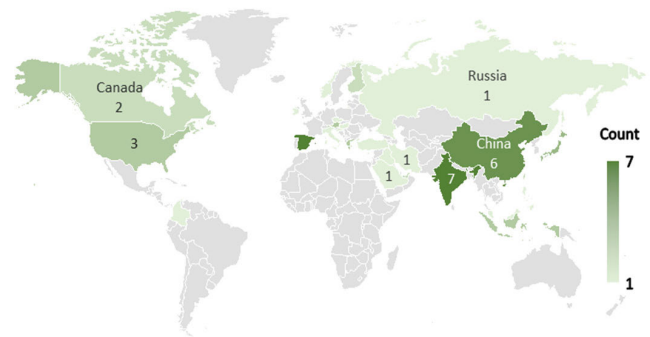


FIGURE 17. Publications origin.

against real-world scenarios, as opposed to simulated ones, suggesting that data from the real world may be more reliable and relevant for the studied context. On the other hand, the most significant number of references for the simulation category is 7. This distribution emphasizes that a wide variety of modeling and simulation techniques exist, but none is as dominant as testing in the real world. The results indicate that while simulations are helpful, they have not yet achieved the same level of validity and generalizability as physical testing in this application.

The visualization in Fig. 18 effectively categorizes the various types of computations according to the number of references they involve. It is interesting to see that "On-board computing" contributes 73% to the entire dataset. This observation indicates a high preference or dependency on on-board computation solutions in the studied case. On the other hand, the remaining computational types (PC computing and Cloud computing) account for only 9% to 18%.

■ On-board processing ■ Cloud computing ■ PC processing

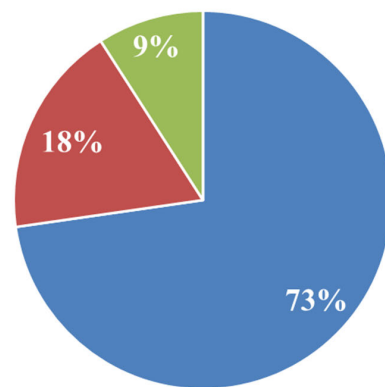


FIGURE 18. Usage proportions of computations type in multi-robot SAR systems.

Onboard processing has become the leading computational paradigm in RSAR systems due to its capability to support low-latency decision-making, ensure user and system autonomy in communication-denied environments, and minimize energy consumption for embedded platforms. Products like the NVIDIA Jetson Nano and Raspberry Pi 4 embody

this trend, with sufficient throughput to run lightweight AI models as well as process sensor data in real-time, right at the edge. In addition, cloud-driven processing environments, although they can often serve computationally heavy intelligence extraction and large data fusion, are by nature deeply dependent on robust, high-bandwidth network infrastructures, which are typically unavailable or unstable in post-disaster settings. Hence, edge computing architectures can be a good compromise between computational efficiency and resilience, opening up the possibility of designing efficient RSAR performance in scenarios where connectivity is limited or infrastructure is degraded.

The utilization rate of different types of robots, including UAVs, UGVs, and UUVs, in robot-oriented SAR systems is summarized in Table 5. Furthermore, the system utilizes multiple sensor types, including thermal cameras, LiDAR, a monocular camera, and an RGB camera. These sensors belong to the robots referenced above. It is interesting to note that aerial robots employ more sophisticated and multimodal sensing solutions than their terrestrial counterparts.

This is mainly because UAVs require real-time awareness of their surroundings and an enhanced ability to perceive their environment to search complex 3D spaces effectively. Heat-sensing cameras, for example, are commonly used in UAVs and can be employed to detect the heat signature of survivors when visibility is poor. High-resolution visual data from RGB and monocular cameras is crucial for tasks such as navigation and object detection in structured and semi-structured environments. On the other hand, the use of LiDAR and multispectral sensors has not been widely adopted yet due to the on-board payload and power consumption limitations of lightweight robots. On the whole, these data also indicate a clear direction towards capitalizing on aerial mobility and multi-sensor principles to increase RSAR coverage and responsiveness further. However, they call for more holistic and energy-frugal data fusion schemes in future systems.

Conversely, the fusion of sensors for robotic platforms in single-robot SAR systems is essential for enhancing SA, autonomy control, and target detection capabilities. Unlike multi-robot SAR systems, single-robot systems must rely on their logarithmically limited onboard sensing and computational resources to accomplish mission-critical tasks in uncertain or dangerous environments. In particular, the results in Table 6 clearly show that the robot body and sensor configurations are chosen according to the specific mission targets and environmental constraints. On the other hand, both thermal cameras and RGB cameras are widely used on single-robot SAR systems with UAVs. More sophisticated versions include monocular vision or LiDAR. This high prevalence is due mainly to the capability of UAVs for quickly exploring large and hard-to-reach terrains, which also makes them highly effective for post-disaster reconnaissance, victim detection, and situation mapping. Infrared cameras can detect the presence of people based on their infrared signatures, especially in dark or occluded environments. Meanwhile, RGB and LiDAR sensors enable accurate obstacle avoidance,

terrain mapping, and visual recognition. The examination demonstrates a clear focus on improving sensor utilization through onboard data fusion to mitigate the lack of cooperative entities. In general, the trends in Table 6 illustrate the increasing degrees of autonomy present in single-robot SAR systems, as well as the current weaknesses in sensor payload integration and on-board processing efficiency that warrant further investigation.

IV. DISCUSSION

This section critically examines existing gaps in robotic SAR technologies, delineates the methodological constraints inherent to this systematic review, and articulates prospective research directions aimed at advancing autonomous functionality, multi-agent coordination, and practical field deployment in real-world disaster response scenarios.

A. GAP ANALYSIS

To systematically analyze these challenges, we classify the gaps into six primary categories:

1) MULTI-ROBOT COORDINATION IN DYNAMIC ENVIRONMENTS

To date, there is a lack of efficient multirobot coordination techniques for SAR in dynamic and uncertain environments (e.g., collapsed buildings and flood areas). The predominant current approaches involve central organization, but this is unreliable in disaster zones, where it can block or damage infrastructure. This approach resulted in inefficient search coverage, increased response times, and a lack of capability to dynamically reassign tasks among the robots.

2) REAL-TIME AI-DRIVEN DATA PROCESSING AND DECISION-MAKING

Nowadays, AI-based detection of victims, as well as path-planning algorithms, suffer from the problem that they are not suitable for real-time adaptation due to limited computational resources. Particularly, AI models for survivor detection often struggle to achieve high accuracy under low-visibility or deteriorated sensor conditions (e.g., smoke and underwater environments). False positive/negative rates in detecting survivors will increase, and both navigation and obstacle avoidance efficiency of the quadrotor will be affected.

3) SENSOR FUSION AND ENVIRONMENTAL PERCEPTION

Most current robotic-based SAR systems are sensor-specific and cannot be readily adapted for use under different environmental conditions. Thermal cameras, for example, do not function well in high-heat environments, and RGB cameras do not perform well in low-light situations or when there is dust present. Multimodal sensor fusion can be enhanced by incorporating LiDAR, thermal imaging, hyperspectral imaging, and biosensors to improve perception robustness in diverse environments.

TABLE 5. Multi-robot SAR system: previous study comparison.

Ref	Path Planning	Computational type	Testing environment	Sensors	Robot-type	Detection algorithm
[28]	DARP algorithm	Cloud Computing	Simulation	• Thermal Camera and Audio.	UAV	CNN algorithm
[30]	Adaptive Flight Path	On-board computing	Simulation	• RGB and Thermal camera, Marine Radar, LiDAR, and • Weather sensors.	UAV, boat	Lightweight Edge AI model
[33]	flexible and adaptive path planning	On-board computing	Simulation	• Camera	UAV	NA
[37]	Footstep navigation trajectory	On-board computing	Real-world	• Multi-stereo camera, • LiDAR	DJI Matrice 100 and a humanoid robot	Darknet algorithm
[48]	Autonomous movement	On-board computing	Real-world	• Thermal camera	E-PUCK2 robot	SVM with linear kernel
[49]	Manual control	PC computing	Real-world	• Thermal camera, RGB camera	UAV	AOS algorithm
[56]	Manual control	On-board computing	Real-world	• UAV-camera, Bio-Radar Senso	Swarm of UAVs	YOLOv4-Tiny
[57]	SO-DRL algorithm	On-board computing	Real-world	• LiDAR	UAV, UGV	NA
[64]	Centralized control system with a "leader" quadcopter	On-board computing	Real-world	• Depth camera sensor	Quadcopter	YOLO Algorithm
[65]	Centralized Approach	Cloud computing	Simulation (MASPlanes++)	• RGB-D camera	UAVs	NA
[74]	Autonomous swarm navigation	NA	Simulation (V-Rep)	• RGB camera	UAVs	M-EC algorithm
[88]	DEM method	NA	Real-world	• monocular camera	Multiple DJI	YOLOv5
[43]	Autonomous flocking behavior	On-board computing	Real-world and Simulation (Gazebo)	• RGB Camera, UV Cameras, UV LED, Rangefinder	UAVs	CNN "person detector algorithm
[43]	NA	On-board computing	Real-world (indoor)	• RGB camera	NA	HHD technique

4) LIMITED ENERGY AUTONOMY AND EXTENDED MISSION DURATIONS

Most SAR robots, particularly UAVs, suffer from limited battery life, significantly restricting their operational time in large-scale disaster scenarios. These robots require frequent recharging, or battery replacement disrupts search operations and delays victim retrieval.

5) ENHANCING HUMAN-ROBOT INTERACTION FOR EFFECTIVE MISSION CONTROL

The ongoing problem is that traditional HRI interfaces have not been adequately refined for real-time work under pressure in an SAR environment. There has been limited development in the field of gesture-based, voice-controlled, and BCI for RSAR systems. This struggle places an additional cognitive burden on SAR members, slows decision-making, and can result in errors.

6) ADAPTIVE MISSION STRATEGIES FOR HETEROGENEOUS ROBOT TEAMS

Many works on SAR assume uniform robotic swarms; however, in reality, we need heterogeneous teams (including UAVs, UGVs, UUVs, or humanoids) that can work together effectively. The newer challenge, which is the subject of this paper, is the lack of standard frameworks for interoperability, resulting in improper task executions. To address this limitation, we need to design an adaptive mission planner that enables different robots to dynamically switch roles according to the environmental constraints (e.g., UAVs perform reconnaissance, UGVs conduct detailed ground searches, and UUVs explore flooded areas).

B. METHODOLOGICAL LIMITATIONS

Although this review was conducted according to established systematic review principles, several methodological and contextual limitations must be acknowledged. These limitations are outlined below to clarify the scope of the findings and guide future researchers in improving the replicability and comprehensiveness of their work.

1) PUBLICATION AND SELECTION BIAS

Despite the inclusion of a structured multi-database search, this review is potentially subject to publication bias. Some pertinent studies, particularly those from industrial or governmental projects, may be disregarded due to the non-indexing of grey literature in academic databases. This is a standard limitation of robotics systematic reviews, as practical yet non-peer-reviewed content is often underrepresented.

2) SUBJECTIVITY IN DATA EXTRACTION

Although a structured review protocol was followed, some degree of subjectivity is inevitable in interpreting and categorizing methodological features, particularly when papers lack standardized terminology or comprehensive technical details.

3) HETEROGENEITY OF STUDIES

The included studies vary significantly in terms of experimental setup, evaluation metrics, disaster scenarios, and sensor-robot configurations. This heterogeneity limits direct comparison and may dilute the generalizability of observed trends.

TABLE 6. Single robot SAR system: Previous study comparisons.

Ref	Path Planning	Computational type	Testing environment	Sensors	Robot-type	Detection algorithm
[89]	predefined grid movement	NA	Real-world	• Monocular camera and ToF sensor	DJI Robomaster	DeepLabv3+
[90]	NA	NA	Real world	• RGB camera	UAVs	RescueNet
[140]	Manual control	PC computing	Real-world	• RGB-D camera and laser sensor	ARTUR	YOLOv5
[91]	Manual control	PC computing	Real-world	• Thermal camera, IR sensors	UGV	YOLO, SSD, Faster R-CNN
[92]	Manual control	On-board computing	Real-world	• Thermal camera	UAV	YOLOv5
[93]	Shortest path planning	On-board computing	Real-world MATLAB SIMULATION	• PIR sensors	UAV	PIR's human detection
[94]	Manual manually or automatically controlling	PC computing	Real-world	• Thermographic	Parrot Bepop2 drone	SSD model
[95]		On-board computing	Real-world	• PIR sensors	UGV	PIR's human detection SSD model
[97]	POMDP algorithm	On-board computing	Real-world	• TFMini • Plus range sensor, T265 sensor, RGB camera	Holybro S500	
[43]	NA	On-board computing	Real-world (indoor)	• RGB camera	NA	HHD technique
[29]	Adaptive Flight Path	On-board computing	Real-world (dense forests)	• Thermal camera	UAV	YOLOv4-tiny
[31]	3D Path Planning	Gourbi V8 MIQP	Simulation	• Onboard Camera • Thermal and IR camera,	UAV	probabilistic detection
[32]	Value Sensitive Design	NA	theoretical only	• LiDAR, • Weather sensors, • Joint Force Sensors • RGB camera, thermal camera	Snake robot	Braitenberg algorithm
[34]	Manual control	On-board computing	Real-world	•	UAV	Yolov4
[35]	Manual control	On-board computing	Real-world	• Thermal Camera	DJI Matrice 210	YOLOV3 and MobileNet
[36]	Autonomous movement and Manual control	On-board computing	Real-world	• LiDAR, RGB-D camera, Thermal Camera,	Mobile robot (karo)	NA
[38]	RRT planner	On-board computing	Real-world	• CO2 sensor • RGB, thermal, and multispectral cameras	Quadrupedal robot	YOLOv8-m
[39]	Manual control	On-board computing	Real-world	• RGB – camera	UAV	YOLOv3
[40]	reinforcement learning-based motion control	On-board computing	Simulation (CoppeliaSim)	• Vision Sensors	snake robot	reinforcement learning-based
[41]	NA	Offline processing	Real-world	• RGB camera	UAV	YOLOv4
[42]	Autonomous movement	On-board computing	Real-world (water)	• GoPro camera	UAV	YOLO V3
[45]	Manual control	real-time by the smartphone	Real-world	• Phone Camera	fixed-wing UAV	SSD topology
[46]	Manual control	NA	Real-world	• RGB Camera	DJI Phantom 4 Pro V2	YOLO v3
[47]	Manual control	NA	Real-world	• Thermal Camera	UAV	Mobile net-SSD
[50]	Manual control	On-board computing	Real-world	• optical Camera, thermal camera	3DR Solo Quadcopter	OpenCV functions
[51]	Manual control	NA	Real-world	• Thermal Camera	Phantom 3 Pro	YOLOV3
[52]	NBV approach	PC computing	Simulation (Gazebo)	• Thermal camera, RGBD camera, wireless transmitter	ground vehicle	SSD, RSSI localization
[53]	Q-value network	NA	Real-world	• Omnidirectional camera, a rotating laser scanner, and a pan-tilt thermal camera	Skid-steer search-and-rescue robot	CNN network
[54]	Manual control	PC computing	Real-world	• Bio-radar, multispectral camera	M100 Quad-rotor	Detection Based on Multisp
[55]	NA	NA	Real-world	• RGB camera	snake-like robot	HVDA approach
[58]	predetermined waypoints	PC computing	Simulation (Gazebo)	• RGB camera	UAV	YOLOv7
[59]	MPC algorithm	NA	Simulation	• RGB camera	UAV	NA
[61]	Manual control	NA	Real-world	RGB camera	DJI Phantom 3 Pro	NA
[60]	Manual control	NA	Real-world	RGB camera	DJI Mavic Pro	NA
[62]	semi-autonomous reactive control	On-board computing	Real-world	RGB-D camera	octocopter	CPM algorithm
[63]	Manual control	PC computing	Real-world	Action camera	UAV	OpenPose approach
[66]	Manual control	PC computing	Real-world	loudspeaker, an array microphone, a camera	UAV	MUSIC, YOLO v3

TABLE 6. (Continued.) Single robot SAR system: Previous study comparisons.

[67]	Manual control	On-board computing	Real-world	RGB camera	X-UAV Talon	Color signature identification
[68]	Manual control	NA	Real-world	RGB camera	UAV	CNN algorithm
[69]	Manual control	Cloud computing	Real-world	OV2640 camera sensor	UGV	NA
[70]	Manual	On-board computing	Real-world	PIR sensor	UGV	PIR human detection
[71]	Automatic waypoint	On-board computing	Real-world	Infrared sensors, Microwave sensor, Thermal sensor	UGV	NA
[72]	Manual control	On-board computing	Real-world	thermal camera, RGB-Depth sensor	Quadruped Robot	YOLOv3
[73]	Semi-autonomous control	On-board computing	Simulation (Gazebo), Real-world	LiDAR, Raspberry Pi Camera,	TurtleBot3 Burger	AprilTag detection
[75]	Automatic and manual control	On-board computing	Real-world	RGB camera, ultrasonic sensors	IROB-IP robot	Face detection by OpenCV
[76]	Indirect search (IS) and next-best view (NBV) techniques	PC computing	Real-world	RGB-D camera and a LiDAR	quadruped robot	YOLOv5
[77]	autonomous control	On-board computing	Real-world	Thermal camera, night vision camera, TF Luna LiDAR, Gas sensor, microphone, and speaker	Snake robot	Thermal sensing and heartbeat detection
[78]	Manual control	On-board computing	Real-world	NIR Camera, Odor Sensor	Quadcopter	MobileNetV2
[79]	NA	NA	Real-world	RGB-D camera	TurtleBot3 Burger	RetinaNet
[80]	RRT algorithm	PC computing	Real-world	LiDAR sensors, Thermal cameras, RGB-D cameras	quadruped robot	CNN algorithm
[81]	Manual control	PC computing	Simulation (ROS/ Gazebo)	RealSense camera	quadruped robot	YOLOv8
[82]	Semi-autonomous	On-board computing	Real-world	Thermal camera IP camera, Ultrasonic Sensor	UGV	propose human detection
[83]	Semi-autonomous	On-board computing	Real-world	LiDAR	UGV	GPR algorithm
[84]	Manual control	PC computing	Real-world	microphones	UAV	Sound human detection
[85]	Manual control by joystick	On-board computing	Real-world	Optical camera	Mobile robot	SSD MobileNet V1
[86]	Manual control	PC computing	Real-world	microphones, speakers, vibration sensors, and a high-speed camera	Active Scope Camera (ASC) robot	survivor's voice detection
[98]	Manual control	On-board computing	Simulation	Thermal sensor, RGB camera	SnakeBotSaver	Detecting human body heat
[87]	D* algorithm	PC computing	Simulation (Pyrosim)	LiDAR and thermal cameras	UGV	SSDA method

4) LACK OF QUANTITATIVE META-ANALYSIS

Due to the descriptive and diverse nature of the included studies, no quantitative synthesis (e.g., meta-analysis) was performed, which limits the ability to statistically assess effect sizes or performance differences across platforms or sensor types.

C. FUTURE DIRECTIONS

To address these gaps, future research should focus on:

1. Designing bio-inspired and energy-efficient locomotion for long-term SAR.
2. Considering decentralized multi-agent robotic system for robust task allocation based on AI techniques.
3. Developing methods for real-time sensor fusion for enhanced situational awareness.
4. Improving HRI systems using AR, BCI, and gesture-based control for more effective HRI in SAR.
5. Standardizing multi-robot interoperability protocols to better coordinate heterogeneous fleets.

V. CONCLUSION

This review systematically examined the robotic SAR systems and their contributions to improving search capability and reducing mission risk. The paper classified SAR robots according to their platform type and examined the AI, sensor technologies, and human-robot interaction features incorporated into these robots.

The results demonstrated that, when compared to UAVs, multi-robot systems combining aerial, ground, and marine robots exhibit notable superiority in complex SAR environments. One significant contribution of this work is the development and analysis of a multi-layer architecture for an SAR system, which increases system modularity while achieving higher efficiency. The HRI Layer enables efficient cooperation between human operators and robots, supporting integrated control and decision-making. The Perception Layer incorporates a wide range of sensors, including LiDAR, thermal cameras, and biological sensors, to enhance environmental perception and victim detection.

Electronically, the Navigation Layer enables autonomous movement through path planning algorithms, obstacle avoidance, and multi-robot coordination. The Control Layer (responsible for real-time system control and motor commands) and the Robot Layer are composed of physical robotic platforms performing SAR missions. This organized manner allows real-time responding in disaster areas as well as collaboration among multiple agents. In practice, the challenges remain in real-time AI-enabled data analysis, energy management, and robust communication network protocols for multi-robot coordination, despite these developments. Accordingly, future work should focus on enabling autonomous capabilities based on AI-driven decision-making, maximizing energy efficiency for extended missions, and promoting cooperation between robotics and humans in SAR operations. In particular, overcoming these gaps will lead to continued improvements in the performance of RSAR, and these robots will likely be better equipped to respond to real-world disasters.

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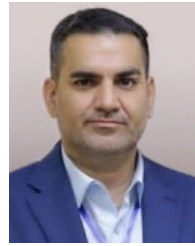
ABBAS AQEEL KAREEM received the B.Sc. degree in computer engineering techniques from Dijlah University College, Baghdad, Iraq, in 2019, and the M.Sc. degree in computer engineering techniques from Middle Technical University (MTU), Baghdad, in 2022, where he is currently pursuing the Ph.D. degree in computer engineering.

He contributed to organizing the 2024 AIP Conference, held at the Electrical Engineering Technical College, MTU. He is an Academic Staff Member with MTU. His work involves various technologies and methods, including AI techniques, image processing, drone systems, and PID controller design. His research interests include unmanned aerial vehicles (UAVs), artificial intelligence (AI), and optimization algorithms.



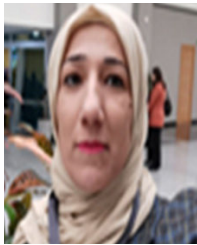
technologies, power electronics, and energy conservation, with particular emphasis on enhancing the reliability of PV systems through intelligent diagnostic frameworks. His research interests include fault detection in photovoltaic (PV) systems, the industrial Internet of Things (IoT), solar tracking, and energy management systems.

AHMED JABBAR ABID received the Ph.D. degree in electronics and communications engineering from the University of Basrah. He is currently the Head of the Space and UAV Engineering Techniques Department and a Senior Faculty Member with Middle Technical University, Baghdad, Iraq. He has published more than 60 research papers in peer-reviewed journals and conference proceedings. His current work focuses on bio-inspired optimization, renewable energy



He is currently a Professor with the Department of Medical Instrumentation Techniques Engineering and holds the position of the Vice-Dean for administrative and financial affairs at the Electrical Engineering Technical College. He is also affiliated with the School of Engineering, University of South Australia, Mawson Lakes, SA, Australia, serving as an Adjunct Associate Professor, since 2022. His research interests include biomedical instrumentation and sensors, health-care applications, computer vision systems, and microcontroller applications. He is an Engineers Australia (EA) Member, in 2018, and an International Association of Engineers (IAENG) Member, in 2018.

ALI ABDULELAH AL-NAJI (Member, IEEE) received the Bachelor of Engineering degree in medical instrumentation techniques from the Electrical Engineering Technical College, Middle Technical University, Baghdad, Iraq, in 2005, the master's degree in electrical and electronic engineering from the University of Technology, Baghdad, in 2008, and the Ph.D. degree in electrical and information engineering from the University of South Australia (UniSA), Australia, in 2018.



Ukraine) and published in leading journals, such as IEEE ACCESS, *Cogent* (MDPI), *Sensors*, *Future Internet*, and Springer. Her research focuses on computer network security, cryptography, wireless sensor and body area networks, and image processing, using AI techniques like neural networks, genetic algorithms, PSO, memetic algorithms, game theory, and optimization methods.

DALAL ABDULMOHSIN HAMMOOD (Member, IEEE) received the B.Sc. degree in computer engineering and IT from the University of Technology, Baghdad, in 2003, the M.Sc. degree in computer engineering techniques from Middle Technical University, in 2011, and the Ph.D. degree from Universiti Malaysia Perlis (UniMAP), in 2020. She is currently an Assistant Professor. She has contributed to several IEEE and international conferences (Malaysia, Egypt, Jordan, and



encouraged by other researchers more than 5 000 times. His research provides ample evidence of his breadth of expertise, including avionics for drones, medical technology, space exploration, neuroscience, and behavioural biology. High points of his career have included working with NASA Ames, developing Mars exploration technology, and supporting his military colleagues. His research is multidisciplinary, collaborative, and encompasses both theory and application. As a Research Degree Supervisor, he is highly sought after as a supervisor and mentor and is responsible for a large and diverse group of graduate students. His research students have gone on to attain senior academic and government research positions in Australia and overseas. He encourages collaboration, and communications, with a focus on innovation, new products, and new applications.

JAVAN CHAHL received the Ph.D. degree in neuroscience and robotics from Australian National University. He is currently the DST Group Joint Chair of Sensor Systems and has been with UniSA, since 2012. Prior to this, he was a Professor of aerospace with RMIT University, and before that, he was with the Defence Science and Technology (DST) Group, with whom he still works closely. He has published more than 200 full-length refereed articles and has been ref-

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